Application Note

NovaSource G6™

RF Signal Source

Programming the *NovaSource G6*

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All *NovaSource G6* models use a simple serial interface that is compatible with industry standard PC COM ports. This allows control of the *G6* models from any PC terminal program such as ProComm or HyperTerm or directly from any UART, such as is common on many embedded processors and microcontrollers.

This application note describes the hardware and command set needed to control the *NovaSource G6* and describes several applications of it, such as filter characterizations and bit error rate testing.

Hardware Required

To control the *NovaSource G6* from a computer, a standard COM port is required along with a standard serial port cable with DB9 connectors at each end. A variety of standard cables are available from different vendors, such as part number AK131-2 from Assmann.

The *NovaSource G6* is wired as DTE so that a null modem is not needed when using it with a standard PC. The connection diagram is shown below and conforms to the industry standard pinout for DTE using a DB9 connector.

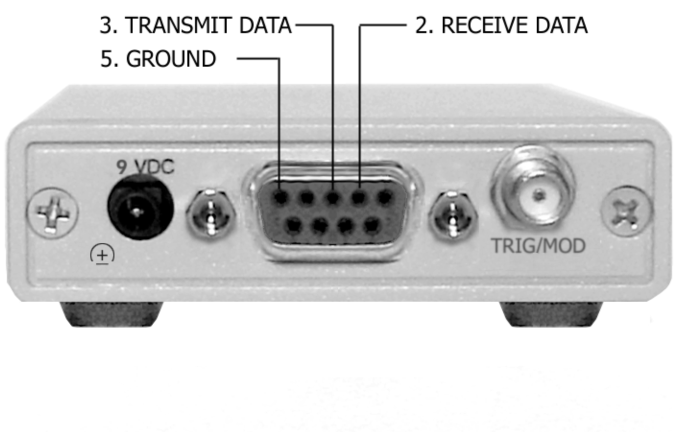

Figure 1: *NovaSource G6* DB9 Pinout Diagram

Serial Port Settings

The host serial port should be set as follows:

- Baud Rate: 38400 bps
- Data bits: 8
- Parity: None
- Stop Bits: 1

No handshaking is needed. Signal lines should rest in the mark state when no data is being transmitted. Data is sent least significant bit first.

Signaling is at true RS232 levels, so that embedded applications will need a level translator to achieve ±3 to ±25 volts levels.

When using a terminal program to control the *NovaSource G6*, the program should be set to send a line feed with the carriage return as the end-of-line character. Similarly, the program should be set to translate an incoming carriage return to carriage return with a line feed. The local echo should be enabled since the *NovaSource G6* does not echo each character.

Proper operation of the serial port connection can be easily verified by applying power to the *NovaSource G6* after the serial connection is ready. At power-up, the *NovaSource G6* sends out the following prompt:

NS G6 v1.0 ${}^{C}{}_{R}$ (c) 2002 Nova Engineering, Inc. ${}^{C}{}_{R}$ >

If this prompt does not display in the terminal window, check for proper connection and settings, and verify the integrity of the PC com port.

NovaSource G6 interprets the backspace character (hexadecimal 0x08) as a destructive backspace so typing errors can be corrected while using the terminal program.

Command Protocol

All commands are entered as standard ASCII characters. Characters may be upper or lower case. All characters in responses from the *NovaSource G6* will be upper case. Commands and responses are all terminated by a carriage return (hexadecimal 0x0D) with no line feed required. A quick reference page is included at the back of this manual.

Each command will result in a response from the *NovaSource G6* with the response content dependent on the command that was issued, either a Set or a Query.

Note: In the following text, $^{C}{}_{R}$ denotes the carriage return character, hexadecimal 0x0D.

Sets

Issuing a command with a parameter value causes the *NovaSource G6* to set the parameter to that value and results in a response of “ok” with a prompt. Example:

Command: AT15 $^{C}{}_{R}$ Response: OK $^{C}{}_{R}$ >

In this example, the *NovaSource G6* is commanded to set the internal attenuator to step 15 and the *NovaSource G6* echoes the command, indicating it has successfully carried out the command.

Queries

Issuing a command with no parameter returns an echo of the command with the value of the relevant parameter. Example:

Command: AT$^{C}_{R}$ Response: OK 21 $^{C}_{R}$ >

In this example, issuing the Attenuation Control command without a parameter will return the current setting of the attenuation control in the attached *NovaSource G6*.

Error Conditions

Commands issued to the *NovaSource G6* may be invalid for several reasons. If this happens, the *NovaSource G6* will indicate the reason for the failure of the command. Note that in the case where no delimiter is sent (a carriage return), the *NovaSource G6* will not respond at all because it is looking for the carriage return that indicates the entire command has been received. If no carriage return is received, the *NovaSource G6* continues to monitor the incoming data without acting on any commands.

Commands may also be invalid because the command itself is not recognized. If there are typos in the command letters, the issued command will not match any known command and will be rejected. When a carriage return is received the *NovaSource G6* will parse the command and fail to find a match in its dictionary. This will be indicated by a response from the *NovaSource G6* of

ER 1 $^{C}_{R}$

A second condition that will cause commands to be rejected is if the parameter value is outside of the legal values for the given command, such as when attempting to program the *NovaSource G6* frequency to a value outside the frequency range of the unit. In this case the *NovaSource G6* will respond with

ER 2 $^{C}_{R}$

Other error conditions that may occur that do not fall under the previous cases will be indicated by a response from the *NovaSource G6* of

ER 3 $^{C}{}_{R}$

Command Set

Notation

In the following command descriptions, parameters are indicated by c, i, or f. A space can be sent between the command letters and the parameter value, but it is not required.

c indicates a single character is used as the parameter. For example, ETc means enter ET followed by a single character such as

ETD$^{C}{}_{R}$

f indicates that a floating point value is intended. This is used for commands involving frequencies. For example, FRf means enter FR followed by a numeric value including a decimal point, such as

FR902.15$^{C}{}_{R}$

i indicates that an integer value, a numeric value with no decimal point, is intended. For some commands the allowed range may include negative numbers, indicated by a minus sign preceding the value. A plus sign is optional when entering positive integers. For example, to set the attenuation control to the 25th step, enter

AT25$^{C}{}_{R}$

Attenuation Control	Ati	i= 0 to 31

Sets the attenuation control to the step indicated. This is approximately equal to the attenuation in dB from the maximum output available from the *NovaSource G6*. Therefore, AT0 $^{C}_{R}$ sets the *NovaSource G6* to the maximum RF output level.

Internal Trigger	ITc	c = E or D

To assert the internal trigger (thereby **E**nabling the RF output from the *NovaSource G6*), enter ITE. To de-assert the trigger (**D**isabling the RF output) enter ITD. Note that the internal trigger and external trigger can not be used at the same time. To make the internal trigger the active trigger, the external input must be set for modulation (IMM).

Frequency	FRf	f = 0.0 to 5875.000

Use this command to tune the *NovaSource G6* to the desired frequency. The allowed range of values of f is limited to the frequency range (in MHz) supported by the attached *NovaSource G6*. For example, for the NS3-2400102, f can be any value between 2400.000 and 2900.000 MHz. The *NovaSource G6* can be tuned with 1 kHz resolution. Leading zeroes and trailing zeroes are not required.

Input Mode	IMc	c = T or M

This command selects the function of the Trigger/Modulation connector on the *NovaSource G6* rear panel. To supply a **T**rigger signal to the *NovaSource G6* enter IMT. To provide **M**odulation to the *NovaSource G6* enter IMM.

Load	LD

The load command requires no parameters. Load will cause the *NovaSource G6* to read the last stored settings from its internal non-volatile memory and make them the active settings. This can be used to restore the *NovaSource G6* to a known state or to simplify repetitive tasks that always start from the same settings. To save the current settings to memory, refer to the STore command.

LED State	LS

The LED State command is a <u>read only</u> command that returns the state of the front panel LEDs on the *NovaSource G6*. The response will be LS followed by a single byte that contains three bits defining the LED states as follows:

Bit 0 = Power LED (Should always report as 1 = on)

Bit 1 = Lock LED (1 = output is locked, 0 = not locked)

Bit 2 = RF State (1 = RF on, 0 = RF off)

Bits 3 – 7 = unused, no meaning, always read as 00110

This command can be used by software applications to determine the RF output state or lock condition of the *NovaSource G6*. The ASCII character displays as a number 1 through 7, and should be interpreted as shown in the table below.

Table 1: ASCII Character Reference

Number	Bits 0-2	RF State	Locked	Power
1	001	OFF	NO	ON
3	011	OFF	YES	ON
5	101	ON	NO	ON
7	111	ON	YES	ON

Modulation Gain	MGi	i = -10 to +13

This command can be used to change the amplitude of deviation when using the FM modulation function. The gain value affects both internal and external modulation, and multiplies the incoming signal by two raised to the selected value. For example, the command “MG-3 $^{C}_{R}$“ would multiply the incoming modulation signal by a factor of 2^{-3} or 0.125. Therefore, a higher gain value will increase the FM deviation. For positive values, the “+” is optional.

Modulation Source	MSc	c = I, E, or N

This command controls the modulation function of the *NovaSource G6*. To enable the use of an **E**xternal modulation signal, enter MSE. After MSE is received by the *NovaSource G6*, any voltage applied to the external input will modulate the RF output. To enable the **I**nternal 1 kHz modulation source, enter MSI. To turn off all modulation, enter

MSN. When modulation is turned off, the RF output will return to the programmed center frequency.

Reference Frequency	RFf	f = 0.0 to 50.000

Sets the reference frequency (in MHz) that the *NovaSource G6* will use in tuning the internal PLL. This value is used in internal calculations when tuning the output to a particular frequency. If the value programmed does not match the actual reference frequency, undesirable operation may result. The default and internal reference are 10.000 MHz.

RF Standby Mode	RSc	c = E or D

If RF Standby mode is **E**nabled the RF output can be turned on and off quickly. **D**isabling RF standby mode will produce a lower signal level when the output is turned off, but will require a slightly longer time to turn on and off. See the *NovaSource G6* manual for a complete description of RF standby mode.

Store	ST

The counterpart to the Load command, Store causes the current settings to be written to the non-volatile memory within the *NovaSource G6*. These settings will then be recalled at power up as the default settings.

Trigger Mode	TMc	c = C, M, or T

When a trigger is received, the *NovaSource G6* will interpret it as **C**ontinuous, **M**omentary, or **T**oggle. This command applies to the active trigger source, whether the external input or the serial command. Refer to the *NovaSource G6* user manual for a complete description of Trigger modes.

Low Frequency	LF

This command is read only and returns the lowest frequency that the unit can support. The LF command is intended as a means of identifying models in applications where *NovaSources* are changed frequently.

High Frequency	HF

This command is read only and returns the highest frequency that the unit can support. The HF command is intended as a means of identifying models in applications where *NovaSources* are changed frequently.

Sample Application 1: Filter Characterization

Filters are typically evaluated using network analyzers costing thousands of dollars. These bulky, expensive tools can be replaced using the *NovaSource G6* and a standard spectrum analyzer. In a network analyzer, a signal source is applied to the filter input and is swept across the frequency band of interest while a spectrum analyzer measures the output power from the filter output.

Sweeps between two frequencies can be easily generated using the *NovaSource G6* by commanding it to each of the target frequencies. A series of frequency commands, FR, is issued that tune the *NovaSource G6* to each frequency, in sequence. The step size between frequencies can be as small as 1 kHz and the band of interest can be as wide as the entire tuning range of the *NovaSource G6*.

A spectrum analyzer connected to the filter output will then display the output level from the filter as the *NovaSource G6* sweeps across the band. Adjusting the video bandwidth and display persistence for the best display will provide a usable filter response picture.

Most terminal programs support some type of script or macro language. Simple scripts can be used to automate the series of FR commands. Consult the manual for your particular program to learn how to use these scripts. In addition, software packages such as Microsoft's Visual Basic™ provide a simple way to access the COM port and write an application to step the *NovaSource G6* through the frequency range. This also provides great flexibility to design other applications for the *NovaSource G6*.

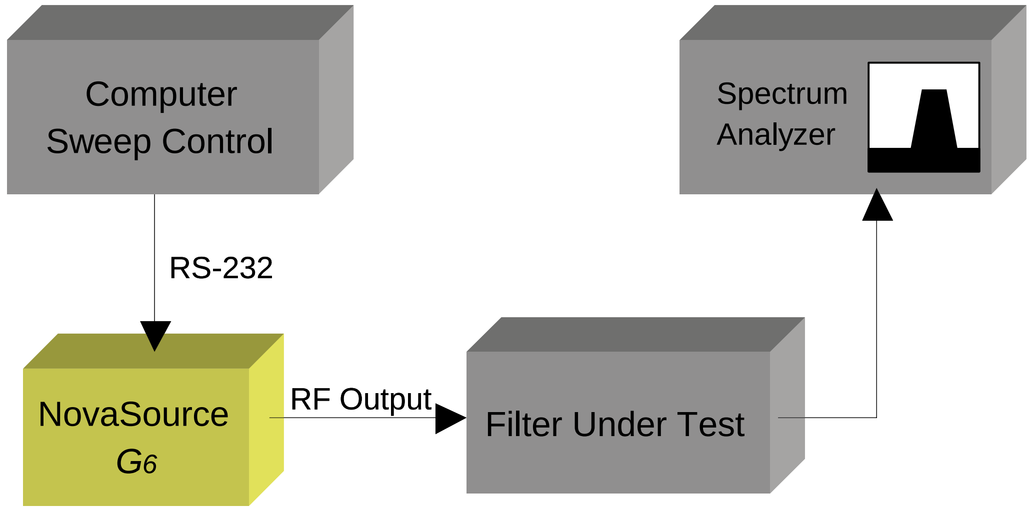

Figure 2: Sweep Generator Flowchart

In order to determine such filter characteristics as insertion loss, it is necessary to know the power level of the signal entering the filter. Since the attenuation control of the *NovaSource G6* is not calibrated to an absolute level, it is necessary to measure the output power using the spectrum analyzer. To do this, connect the *NovaSource G6* directly to the spectrum analyzer and adjust the reference level so that the peak of the signal is just at the top of the screen, or at some other fixed level. Reconnect the *NovaSource G6* to the filter input, and the filter output to the spectrum analyzer. The difference between the original signal level and the filter output level is the insertion loss.

Sample Application 2: Receiver Bit Error Rate Testing

During design of data radios, it is usually necessary to perform a Bit Error Rate test on the RF receiver to determine if the design meets the requirements for fidelity in data demodulation. Normally, this requires a large, expensive piece of equipment to generate a modulated signal that is injected into the front end of the receiver. The demodulated data stream is then monitored and compared to the known correct stream. The number of bit errors, usually per million bits, is the Bit Error Rate, or BER.

The *NovaSource G6* can be used as the modulation source, reducing the cost of a BER test set significantly. This also reduces the space required, even making the test set portable, should that be necessary.

Any standard baseband BER Tester (BERT) can be used to generate a data stream, such as a PN sequence. As shown in the diagram below, the data output of the BERT is supplied to the modulation input of the *NovaSource G6*. The *NovaSource G6* is configured to accept modulation from the external input and is tuned to the desired input frequency of the receiver under test.

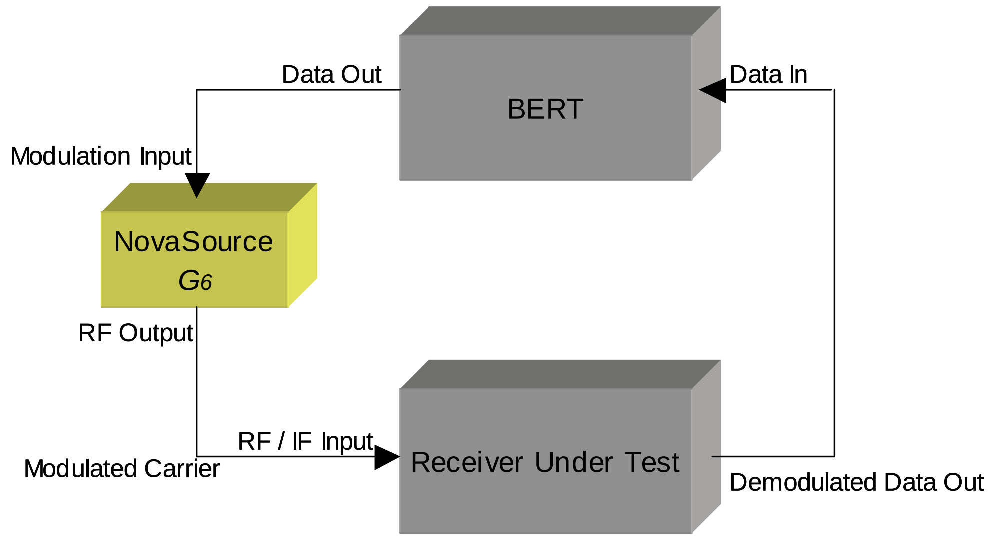

Figure 3: BER Test Set Up

The data output of the receiver is then connected back to the BERT data input, completing the loop. The BERT will monitor the incoming data stream, matching it to the outgoing stream and counting the errors**.**

Table 2: *NovaSource G6* Quick Reference Table

Command	Parameter Range	Function
AT	0 to 31	Attenuation
FR	0.0 to 5875.000	Frequency
HF	None	High Frequency
IM	T, M	Input Mode
IT	E, D	Internal Trigger Mode
LD	None	Load
LF	None	Low Frequency
LS	None	LED State
MG	-10 to +13	Modulation Gain
MS	E, I, N	Modulation Source
RF	0.0 to 50.000	Reference Frequency
RS	E, D	RF Standby
ST	None	Store
TM	C, M, T	Trigger Mode

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Part #: NL-NS002-020315 Programming the *NovaSource G6*

Revision: 2.0